MIL-T-18755B(OS)
16 May 1980
SUPERSEDING
MIL-T-18755A(OS)
10 January 1974
and
AMENDMENT 1
10 December 1974

MILITARY SPECIFICATION

TRACER, MARK 11 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Tracer, Mark 11 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. In event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1370

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	Sampling Procedures and Table for Inspection by Variables for Percent Defective
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition
DOD-STD-2101	Classification of Characteristics

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident. 53711)

DL 394440	Tracer, Mark 11 Mod 0
DL 2128433	Pack, Shipping and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Item definition. This prime item is the Mark 11 Mod 0 Tracer. The tracer is approximately 1.72 inches long and .750 inch in diameter. The tracer weighs approximately 95 grams. The explosive weight of the tracer is approximately 6 grams. The tracer is designed to produce a visible trace of light upon functioning.

3.2 Classification of requirements. The requirements for the tracer are classified herein as follows:

REQUIREMENTS	PARAGRAPH
First article sample	3.3
Design and construction	3.4
Performance and product characteristics	3.5
Workmanship	3.6

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3.3 First article sample. Unless otherwise specified in the contract (see 6.2), a first article sample of tracers shall be furnished by the contractor. The tracers, their components and all material shall be of a single type, grade, class, size, and composition which comply with all applicable provisions of this document. The sample shall be manufactured at the same location using the proposed full production methods.

3.4 Design and construction.

3.4.1 Prime item documentation. The tracer shall be manufactured, examined, and tested in accordance with contract documents, DL 394440, and this specification.

3.4.2 Classification of characteristics. Characteristics verified by the inspections and tests specified herein are classified as Critical, Major and Minor in accordance with DOD-STD-2101. Critical characteristics are identified by the symbol (C), and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Characterisitcs that are not annotated by the classification code symbol are classified as Minor.

3.5 Performance and product characteristics.

3.5.1 Performance.

3.5.1.1 Ballistic functioning. The tracer shall meet the following requirements, when tested in accordance with 4.5.1.1:

a. (C1) Projectile shall not burst in gun bore or in flight within 400 yards of muzzle.

b. (M101) Function (produce a visible trace of light for a minimum period of 7.0 seconds).

c. (M102) Self-destruct.

d. Burning time - minimum 7.4 seconds.

3.6 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents employed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection requirements. The inspection requirements are specified by the classfied characteristics in the specification and on the referenced drawings, and also by the supplemental quality assurance provisions of the contract.

4.1.2 Inspection review. Unless otherwise specified in the contract, the contractor and all test activities identified in the contract shall submit to the Technical Activity and Design Agent for review, a description of the proposed procedure and equipment to be used to perform all examinations and tests (see 6.2).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The contractor shall deliver a sample of 200 tracers to the destination specified (see 6.2), for examination, testing, evaluation, and acceptance as specified in TABLE I, as applicable. Any production of the item prior to authorization by the procuring agency shall be at the risk of the contractor. The sample shall be prepared for delivery as specified in Section 5.

4.4 Quality conformance inspection. Unless otherwise specified, an inspection lot shall consist of 10,000 to 150,000 tracers offered for delivery at one time including test samples (see 6.2). Inspection lots shall be inspected as follows:

a. Any change in materials, suppliers or production methods shall be considered sufficient interruption for the beginning of a new inspection lot. Record reason for interruption on the Data Card (see 6.2).

b. Sampling, examination, testing and acceptance of inspection lots shall be performed as specified in TABLE I.

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TABLE I. Inspection plans.

EXAMINATIONS AND TESTS	FIRST ARTICLE SAMPLING PLANS		INSPECTION LOT SAMPLING PLANS	
Test Sequence	Sample Size 4/	Acceptance Criteria 1/2/	Sample Size 4/	Acceptance Criteria 1/2/
Examinations and Workmanship (4.6) (3.6)	200 Tracers	Critical Ac 0 Re 1 Major Ac 5 Re 0 Minor Ac 10 Re 11	See Footnote 3	See Footnote 3
Ballistic Functioning Test (4.5.1.1)	200 Tracers	(a)Ac 0 Re 1 (b&c) Ac 14 Re 15	MIL-STD-105 Level I	(a)Ac 0 Re 1 (b&c) AQL 4.0
		(d) MIL-STD-414 Code Ltr Q AQL 2.5	MIL-STD-414 Level V	(d) AQL 2.5

Notes

1/ Ac = Acceptance number for defectives of specified requirements of Section 3.
Re = Rejection number for defectives of specified requirements of Section 3.
AQL = Acceptance quality levels as specified in percent defective.
2/ Acceptance criteria shall apply to characteristics enclosed in parenthesis as a group.
3/ Quality assurance inspection requirements of the contract apply.
4/ Mixed variables - Attributes Inspection: samples shall consist of the first consecutively tested units, as applicable to the Standard (MIL-STD-105 or MIL-STD-414).

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4.5 Tests. Random samples of tracers shall be selected and tested as specified in 4.3 or 4.4, as applicable.

4.5.1 Performance tests.

4.5.1.1 Ballistic functioning test. The tracer shall be assembled in a 40-mm projectile Mk 2 HE-T and tested from a weapon as designated by the procuring activity (see 6.2). The projectile shall be fired, single action with service charge and at a temperature of 70 ± 5° F. Suitable aids, such as binoculars, should be used to insure visual contact with the tracer. Record all data required by 3.5.1.1 Defectives are tracers failing to meet the requirements of 3.5.1.1.

4.6 Examinations.

4.6.1 Tracer examinations. Examinations shall be made to assure that the tracers meet the requirements of drawing listed on DL 394440 as applicable. Only tracers with no defects shall be used for further testing or returned to the lot.

4.6.2 Packaging, packing and marking examinations. The packaging, packing, and marking shall conform to Section 5 of this specification.

5. PACKAGING

5.1 Packing and marking.

5.1.1 Level A.

5.1.1.1 One hundred and thirty-five (135) tracers shall be packed and marked in accordance with DL 2128433.

5.2 Ammunition data card. Data cards shall be prepared in accordance with MIL-STD-1167 (see 6.2).

5.3 Ammunition lot number. Ammunition lot number shall be in accordance with MIL-STD-1168.

6. NOTES

6.1 Intended use. The tracer is used in 40-mm projectiles to show a bright spot of light so that the trajectory of the projectile may be observed.

6.2 Ordering data. Procurement documents should specify:

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6.2.1 Procurement requirements.

a. Title, number and date of this specification

b. Quantity being ordered

c. First article samples, if applicable, and test facility where sample is to be shipped for evaluation (see 3.3 and 4.3)

d. Inspection lot size if different than specified in 4.4

e. Inspection review, if applicable (see 4.1.2)

f. Test facility where ballistic functioning tests are to be shipped (see 4.5.1.1)

g. Projectile mark and mod used for test (see 4.5.1.1)

h. Weapon used for testing (see 4.5.1.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENT	APPLICABLE DID
4.5.1	Performance Tests	DI-T-4024
4.4.a and 5.2	Ammunition Data Card	DI-E-2001

Test Reports with individual data will be forwarded to the Technical Activity and Design Agent.

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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6.3 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, or Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Defense Acquisition Regulations (DAR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
NAVY - OS

Preparing Activity:
NAVY - OS

Project No. 1370-N120

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-18755B(OS)	950905

3. DOCUMENT TITLE
TRACER, MARK 11 MOD 0

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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